Chapter 2
Relational Data Model

2.6 Data Integrity Constraints:

Types of Data Integrity Constraints:

I/O Constraint – Primary Key, Foreign Key, Unique Key Constraint.
Business Rule Constraint – Not Null and Check Constraint.

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Data Integrity Constraints

- The term data integrity simply means that the data stored in the table is **valid / authentic**.
- There are different types of data integrity, often referred to as constraints.

NOT NULL Constraint

- A NOT NULL constraint means that a data row must have a value for the column specified as NOT NULL.
- A fairly standard practice is to assign each constraint a *unique constraint name*.
- If constraints are not named, then Oracle assigns meaningless system-generated names to each constraint.

Example:

```
Create table employee
(ename VARCHAR2(25)CONSTRAINT nn_ename NOT NULL,
salary number(8,2)CONSTRAINT nn_sal NOT NULL);
```

PRIMARY KEY Constraint

- Each table must normally contain a column or set of columns that uniquely identifies rows of data that are stored in the table.
- This column or set of columns is referred to as the *primary key*.
- If a table requires two or more columns in order to identify each row , the primary key is termed a *composite* primary key.

Example:

```
Create table emp1
(empno number(4) CONSTRAINT pk_emp1 PRIMARY KEY,
ename VARCHAR2(25)CONSTRAINT nn_ename_emp1 NOT
    NULL, salary number(8,2)CONSTRAINT nn_sal_emp1
    NOT NULL);
```

CHECK Constraint

- Sometimes the data values stored in a specific column must fall within some acceptable range of values.
- A CHECK constraint is used to enforce this data limit.

Example:

```
Create table emp2
(empno number(4) CONSTRAINT pk_emp2 PRIMARY KEY,
ename VARCHAR2(25)CONSTRAINT nn_ename NOT NULL,
 salary number(8,2)CONSTRAINT ck_salary
CHECK(salary <= 500000));
```

UNIQUE Constraint

- Sometimes it is necessary to enforce uniqueness for a column value that is not primary key column.
- The *UNIQUE* constraint can be used to enforce this rule and Oracle rejects any rows that violate the constraint.
- For example, No two employees can be assigned same mobile number.

Example:

```
Create table emp3
(empno number(4) CONSTRAINT pk_emp3 PRIMARY KEY,
ename VARCHAR2(25)CONSTRAINT nn_ename NOT NULL,
 sal number(8)CONSTRAINT ck_sal CKECK(sal<= 50000)
 mob_no number(10) CONSTRAINT un_mob_no UNIQUE );
```

FOREIGN Keys

- Normally, data rows in one table are related to data rows in other tables.
- The *department* table will store information about departments within the organization.
- Each department has a unique, 2-digit department number, department name, and location.

Example:

```
CREATE TABLE dept
(deptno NUMBER(2)CONSTRAINT pk_dept PRIMARY KEY,
 dname VARCHAR2(20)CONSTRAINT nn_dname NOT NULL
 location varchar29(10));
```

Identifying FOREIGN Keys

- Normally, in an organization, at a given time, an employee is assigned to a single department.
- In order to link rows in the ***employee*** table to rows in ***dept*** table we need to introduce a new type of constraint, the **FOREIGN KEY** constraint.
- **Foreign keys (FKs)** are columns in one table that reference **primary key (PK)** values in **another** or in the **same table**.
- Lets relate employee rows to the PK column named ***deptno*** in the **dept** table.
- The value stored to the **emp_deptno** column for any given *employee* row must match a value stored in the ***deptno*** column in the **dept** table.

Identifying FOREIGN Keys

```
CREATE TABLE employee
(emp_no number(5)
        CONSTRAINT pk_employee PRIMARY KEY,
 emp_name  VARCHAR2(25)
        CONSTRAINT nn_emp_name NOT NULL,
 emp_dob DATE,
 emp_salary NUMBER(8,2)
        CONSTRAINT ck_emp_salary
        CHECK (emp_salary <= 55000),
 mob_no number(10) CONSTRAINT un_mob_no UNIQUE,
 emp_deptno NUMBER(2),
CONSTRAINT fk_emp_dpt FOREIGN KEY (emp_deptno)
    REFERENCES dept(deptno) ON DELETE SET NULL);
```

MAINTAINING REFERRENTIAL INTEGRITY

- FOREIGN KEY constraints are also referred to as referential integrity constraints, and assist in maintaining database integrity.
- Referential integrity stipulates that values of a foreign key must correspond to values of a primary key in the table that it references. But what happens if the primary key values change or the row that is referenced is deleted?
- Several methods exist to ensure that referential integrity is maintained.
- ON DELETE SET NULL clause can be used to specify that the value of *emp_dpt_number* should be set to null if the referenced *department* row is deleted.
- There are additional referential integrity constraints that can be used to enforce database integrity.

MAINTAINING REFERRENTIAL INTEGRITY

On Update (Delete) Restrict	Any update/delete made to the *department* table that would delete or change a primary key value will be rejected unless no foreign key references that value in the *employee* table. This is the *default* constraint in Oracle.
On Update (Delete) Cascade	Any update/delete made to the *department* table should be cascaded through to the *employee* table.
On Update (Delete) Set Null	Any values that are updated/deleted in the *department* table cause affected columns in the *employee* table to be set to null.

MAINTAINING REFERRENTIAL INTEGRITY

Specify the behavior for child tuples when a parent tuple is modified.

Action to take if referential integrity is violated:

- SET NULL - Child tuples foreign key is set to NULL - Orphans.
- SET DEFAULT - Set the value of the foreign key to some default value.
- CASCADE - Child tuples are updated (or deleted) according to the action take on the parent tuple.

Example - Primary key

- CREATE TABLE order_header (
  order_number NUMBER(10,0) NOT NULL,
  order_date DATE,
  sales_person VARCHAR(25),
  bill_to VARCHAR(35),
  bill_to_address VARCHAR(45),
  bill_to_city VARCHAR(20),
  bill_to_state VARCHAR(2),
  bill_to_zip VARCHAR(10),
  PRIMARY KEY (order_number));

Example - Foreign key

- CREATE TABLE order_items (

order_number NUMBER(10,0) NOT NULL,

line_item NUMBER(4,0) NOT NULL,

part_number VARCHAR(12) NOT NULL,

quantity NUMBER(4,0),

PRIMARY KEY (order_number, line_item),

FORIEGN KEY (order_number)

REFERENCES order_header (order_number),

FOREIGN KEY (part_number)

REFERENCES parts (part_number));

Example
Table Level Constraint

```
CREATE TABLE order_items (
   order_number      NUMBER(10,0) NOT NULL,
   line_item         NUMBER(4,0)  NOT NULL,
   part_number       VARCHAR(12)  NOT NULL,
   quantity          NUMBER(4,0),
   PRIMARY KEY (order_number, line_item),
   FORIEGN KEY (order_number)
      REFERENCES order_header (order_number)
      ON DELETE SET DEFAULT
      ON UPDATE CASCADE,
   FOREIGN KEY (part_number)
      REFERENCES parts (part_number)
);
```

Example
Table Level Constraint with name

```
CREATE TABLE order_header (
order_number NUMBER(10,0) NOT NULL,
order_date DATE,
sales_person VARCHAR(25),
 bill_to VARCHAR(35),
bill_to_address VARCHAR(45),
bill_to_city VARCHAR(20),
bill_to_state VARCHAR(2),
bill_to_zip VARCHAR(10),
CONSTRAINT order_header_pk
PRIMARY KEY (order_number) );
```

Example : Creating Table from another table & Dropping it

- CREATE TABLE emp_department_1 AS SELECT fname, lname, bdate FROM employee WHERE dno = 1 ;
- create table high_pay_emp as

select * from employee where salary > 50000

- Drop table <table_name>
- Drop table high_pay_emp